

Chapter 6

Relational Calculus

Instructor: Vũ Thị Mỹ Hằng, Ph.D. (vtmhang@fit.hcmus.edu.vn)



fit@hcmus

KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN



Content

- ☐ **Introduction**
- ☐ Tuple Relational Calculus (TRC)





Introduction

- ☐ Is the formal query language
- ☐ Introduced by Codd in 1972
- ☐ Based on logical theory
- ☐ Properties
 - ☒ Nonprocedural language – declarative language
 - ☒ Calculus expression specifies *what is to be retrieved* rather than *how to retrieve*.



Logical Expression

- **Logical expression:** a statement that always has a truth value (true or false).
 - It is August now.
 - $1 > 5$ (a false statement).

- **Concepts:**
 - **Variable:** a changing quantity ($x, y, z, \dots$).
 - **Logical operators** ($\neg$: negation, $\Rightarrow$: implication, $\wedge$: and, $\vee$: or).
 - **Quantifiers** ($\exists$: there exists, $\forall$: for all).
 - **Formula:** expressions constructed based on logical expressions.



Logical Formula

□ Some examples:

- $P(t), \neg P(t)$
- $P(t) \wedge Q(t)$
- $P(t) \vee Q(t)$
- $\neg(P(t) \wedge Q(t)) \quad \Leftrightarrow \quad \neg P(t) \vee \neg Q(t)$
- $\neg(P(t) \vee Q(t)) \quad \Leftrightarrow \quad \neg P(t) \wedge \neg Q(t)$
- $(\exists t)(P(t))$
- $\neg(\exists t)(P(t)) \quad \Leftrightarrow \quad (\forall t)(\neg P(t))$
- $(\forall t)(P(t))$
- $\neg(\forall t)(P(t)) \quad \Leftrightarrow \quad (\exists t)(\neg P(t))$



Relational Calculus Classification

- Categories
 - ▣ **Tuple relational calculus – TRC**
 - ▣ Domain relational calculus – DRC

$$\{ t \mid P(t) \}$$


P is a formula that involves t .
 $P(t)$ is either TRUE or FALSE depending on the value of t .



Content

- ☐ Introduction
- ☐ **Tuple Relational Calculus (TRC)**





Tuple relational calculus – TRC

- A simple tuple calculus query is of the form

$$\{ t \mid P(t) \}$$
$$\{ t.A \mid P(t) \}$$

- t is a tuple variable
 - Its value is any individual tuple from a relation
 - $t.A$ is a value of a tuple t at an attribute A
- $|$ (*vertical bar*) is used to divide the query into two parts:
 - P is a conditional expression involving t
 - $P(t)$ has the TRUE or FALSE value depending on t
 - The result is the set of all tuples t that satisfy $P(t)$



Example 1

- Find all lecturers whose salary is greater than 2000.

$$\{ \mathbf{t} \mid \underbrace{\text{GIAOVIEN}(\mathbf{t})}_{P(\mathbf{t})} \wedge \underbrace{\mathbf{t}.\text{LUONG} > 2000}_{Q(\mathbf{t})} \}$$

- **Result:** Set of tuple $\mathbf{t}$ that satisfy both conditions $P(\mathbf{t})$ and $Q(\mathbf{t})$
 - $P(\mathbf{t})$: $\mathbf{t}$ is a tuple belonging to the relation GIAOVIEN and
 - $Q(\mathbf{t})$: the value of the salary attribute, in tuple $\mathbf{t}$ is greater than 2000

Example 2

- Find the lecturer ID and name of lecturers whose salary is greater than 2000.

$$\{ \mathbf{t.MAGV}, \mathbf{t.HOTEN} \mid \underbrace{\text{GIAOVIEN}(t)}_{P(t)} \wedge \underbrace{t.LUONG > 2000}_{Q(t)} \}$$

- Result** : Set of values **MAGV, HOTEN** from tuples **t** satisfying:
 - P(t)**: t is a tuple belonging to the relation GIAOVIEN and
 - Q(t)**: the value of the salary attribute, in tuple t is greater than 2000



Example 3

- Find the lecturer ID, lecturer name, and the department name where the lecturer works.

$\{ \mathbf{t.MAGV}, \mathbf{t.HOTEN}, \mathbf{b.TENBM} \mid \text{GIAOVIEN}(\mathbf{t}) \wedge \text{BOMON}(\mathbf{b}) \wedge \mathbf{t.MABM} = \mathbf{b.MABM} \}$

- Result:** Set of values **MAGV**, **HOTEN** from tuples **t** and **TENBM** from tuples **b** such as:
 - t** is a tuple in the GIAOVIEN relation,
 - b** is a tuple in the BOMON relation, and
 - b** is the department where t works ($\mathbf{t.MABM} = \mathbf{b.MABM}$).



Tuple Variable Classification

□ Free variable

$$\{ t \mid \text{GIAOVIEN}(t) \wedge t.\text{LUONG} > 2000 \}$$

Free variable

□ Bound variable

$$\{ t \mid \text{GIAOVIEN}(t) \wedge (\exists s)(\text{BOMON}(s) \wedge s.\text{MABM} = t.\text{MABM} \wedge s.\text{TENBM} = \text{“Hệ thống thông tin”}) \}$$

Free variable

Bound variable

Example 4

- Find the lecturer ID of lecturers who belongs to the “Information System” department.

$\{ \mathbf{t.MAGV} \mid \text{GIAOVIEN}(\mathbf{t}) \wedge (\exists \mathbf{s})(\text{BOMON}(\mathbf{s}) \wedge$
 $\mathbf{s.TENBM} = \text{'Hệ thống thông tin'} \wedge$
 $\mathbf{s.MABM} = \mathbf{t.MABM}) \}$



MAGV
1
4

GIAOVIEN		
	MAGV	HOTEN
t1	1	Nguyễn Hoài An
t2	2	Trần Trà Hương
t3	3	Nguyễn Nam Sơn
t4	4	Lý Hoàng Hà

BOMON	
MABM	TENBM
HTTT	Hệ thống thông tin
CNPM	Công nghệ phần mềm
MMT	Mạng máy tính



Example 5

- Find the name of lecturers who participate in at least one project or are the head of a department.

$$\text{CARD}(\{ t \mid P(t) \})$$
$$\{ t.\text{HOTEN} \mid \text{GIAOVIEN}(t) \wedge ($$
$$\text{CARD}(\{ s \mid \text{THAMGIADT}(s) \wedge t.\text{MAGV} = s.\text{MAGV} \}) \geq 1 \vee$$
$$\text{CARD}(\{ u \mid \text{BOMON}(u) \wedge t.\text{MAGV} = u.\text{TRUONGBM} \}) = 1$$
$$\} \}$$

Example 6

- Find the ID, name of lecturers who participate in at least one project whose management level is “ĐHQG”.

{ **t**.MAGV, **t**.HOTEN | GIAOVIEN(**t**) ∧

(**∃s**)(DETAI(**s**) ∧ **s**.CAPQL = “ĐHQG” ∧ (**∃u**)(THAMGIADT(**u**)
 ∧ **u**.MADT = **s**.MADT ∧ **t**.MAGV = **u**.MAGV)) }

{ **t**.MAGV, **t**.HOTEN | GIAOVIEN(**t**) ∧

(**∃s**)(**∃u**)(DETAI(**s**) ∧ THAMGIADT(**u**) ∧ **s**.CAPQL = “ĐHQG” ∧
u.MADT = **s**.MADT ∧ **t**.MAGV = **u**.MAGV) }

Example 7

- Find the lecturers (ID, lecturer name) with the highest salary.

$$\{ t.MAGV, t.HOTEN \mid GIAOVIEN(t) \wedge (\forall s)(GIAOVIEN(s) \Rightarrow t.LUONG \geq s.LUONG) \}$$

*For all s , **if** s is a lecturer, **then** the salary of s is less than or equal to the salary of t*

Example 8

□ List the lecturers (MAGV, HOTEN) who participate in all projects.

$$\{ t.MAGV, t.HOTEN \mid GIAOVIEN(t) \wedge (\forall s)(DETAI(s) \Rightarrow (\exists u)(THAMGIADT(u) \wedge u.MADT = s.MADT \wedge t.MAGV = u.MAGV)) \}$$



MAGV	HOTEN
4	Lý Hoàng Hà

GIAOVIEN	
MAGV	HOTEN
1	Nguyễn Hoài An
2	Trần Trà Hương
3	Nguyễn Nam Sơn
4	Lý Hoàng Hà

DETAI	
MADT	TENDT
1	...
2	...
3	...

THAMGIADT	
MAGV	MADT
1	1
2	2
4	1
4	2
4	3

 ~~1~~
 ~~2~~
 ~~3~~
 t4

 s1
 s2
 s3

 u1
 u2
 u3
 u4
 u5

Example 9

- List the heads of departments (MAGV, HOTEN) who participate in all projects.

$$\{ \mathbf{t.MAGV}, \mathbf{t.HOTEN} \mid \text{GIAOVIEN}(\mathbf{t}) \wedge$$

$$(\exists \mathbf{b})(\text{BOMON}(\mathbf{b}) \wedge \mathbf{b.TRUONGBM} = \mathbf{t.MAGV}) \wedge$$

$$(\forall \mathbf{s})(\text{DETAI}(\mathbf{s}) \Rightarrow (\exists \mathbf{u})(\text{THAMGIADT}(\mathbf{u}) \wedge$$

$$\mathbf{u.MADT} = \mathbf{s.MADT} \wedge \mathbf{t.MAGV} = \mathbf{u.MAGV})) \}$$

For every s , if s is a project, then there exists a participation record u such that lecturer t participates in project s .

Example 10

- List the heads of departments (MAGV, HOTEN) who do not participate in at least one project.

$$\{ t.MAGV, t.HOTEN \mid GIAOVIEN(t) \wedge$$

$$(\exists b)(BOMON(b) \wedge b.TRUONGBM = t.MAGV) \wedge$$

$$(\exists s)(DETAI(s) \wedge \neg(\exists u)(THAMGIADT(u) \wedge$$

$$u.MADT = s.MADT \wedge t.MAGV = u.MAGV)) \}$$

There exists a project s such that there does not exist a participation record u showing that lecturer t participates in project s .



Transform

- ☐ (i) $P_1 \wedge P_2 = \neg (\neg P_1 \vee \neg P_2)$
- ☐ (ii) $\forall t \in R (P(t)) = \neg \exists t \in R (\neg P(t))$
- ☐ (iii) $\exists t \in R (P(t)) = \neg \forall t \in R (\neg P(t))$
- ☐ (iv) $P \Rightarrow Q = \neg P \vee Q$



Safe expression

☐ Examine

$$\{ t \mid \neg(t \in \text{EMPLOYEE}) \}$$

☐ Unsafe

- ☒ Many tuples in the universe that are not EMPLOYEE tuples
- ☒ Even though they do not exist in the database
- ☒ The result is infinitely numerous



Safe expression

- Safe expression
 - ▣ Guarantee to yield *a finite number of tuples*

- A formula P is called safe expression
 - ▣ If its resulting values are from the domain of P
 - The domain of a tuple relational calculus expression: $DOM(P)$
 - The set of all values
 - Either appear as constant values in P
 - Or exist in any tuple in the relation referenced in P



Safe expression

□ Example

$\{ t \mid t \in \text{EMPLOYEE} \wedge t.\text{SALARY} > 30000 \}$

- $\text{DOM}(t \in \text{EMPLOYEE} \wedge t.\text{SALARY} > 30000)$
- The set of values
 - Larger than 30000 at the attribute SALARY
 - Other values at the remaining attributes that appear in EMPLOYEE
- Safe expression



Practice

1. List lecturers who manage at least one other lecturer.
2. List projects and the names of project managers that have more than two tasks.
3. List projects that have participation from “Software Engineer” lecturers.
4. List lecturers who participate in all projects having a budget greater than 50.
5. List lecturers who have the highest salary in each department.

